

Seoyoung Lee Ph.D.

MEDICAL PHYSICS | CT IMAGING & DEEP LEARNING

Rochester, MN, USA • lee.seoyoung@mayo.edu • [ether0.github.io](https://github.com/ether0)

Professional Experience

Postdoctoral Research Fellow

CT Clinical Innovation Center, Dept. of Radiology;
Dept. of Artificial Intelligence and Informatics
Mayo Clinic, Rochester, MN, USA

Sep. 2025 – Present

Education

Ph.D. in Nuclear and Quantum Engineering

KAIST, Daejeon, Republic of Korea

Dissertation: "Development of Advanced Digital Breast Tomosynthesis Imaging Techniques Utilizing Machine Learning"

Advisor: Prof. Seungryong Cho

Mar. 2021 – Feb. 2025

M.S. in Nuclear and Quantum Engineering

KAIST, Daejeon, Republic of Korea

Thesis: "Development of Breast Thickness Correction Method and Shape Reconstruction Method in Digital Breast Tomosynthesis"

Advisor: Prof. Seungryong Cho

Mar. 2019 – Feb. 2021

B.S. in Physics;

B.S. in Nuclear and Quantum Engineering (Dual Major)

KAIST, Daejeon, Republic of Korea

Honors: *Magna Cum Laude* (GPA: 3.85/4.3)

Mar. 2015 – Feb. 2019

Research Interests

X-ray imaging • Tomographic image reconstruction • Image processing • Deep learning • Multimodal learning • Breast imaging • Photon-counting CT

Journal Publications

- 2026** [1] Kim, H., **Lee, S.**, Cho, S. (2026), Source-Free Domain Adaptation for Multi-institutional Chest X-Ray Images. *Journal of Applied Clinical Medical Physics* 27(5), e70626. doi: [10.1002/acm2.70626](https://doi.org/10.1002/acm2.70626)
- [2] Hyun, S. Yun, S. **Lee, S.**, Choi, D. & Cho, S. (2026). Diffusion prior-guided implicit neural representation for metal artifact reduction in sparse-view CT reconstruction. *Physics in Medicine & Biology* 70(10), 105007. doi: [10.1088/1361-6560/ae6a5a](https://doi.org/10.1088/1361-6560/ae6a5a)
- [3] Hyun, S., **Lee, S.**, Choi, I., Shin, C. & Cho, S. (2026). X-ray Source Motion Deblurring Framework for Fast Scan in Digital Breast Tomosynthesis. *IEEE Transactions on Biomedical Engineering* [Early Access]. doi: [10.1109/TBME.2026.3688683](https://doi.org/10.1109/TBME.2026.3688683)
- 2025** [4] **Lee, S.**, Hyun, S., Kim, D., Kim, H., Kim, H., Choi, I., Shin, C. & Cho, S. (2025). Can scout scan help choose scanning angle in digital breast tomosynthesis?. *Journal of Medical Imaging* 12(S2), S22016-S22016. doi: [10.1117/1.JMI.12.S2.S22016](https://doi.org/10.1117/1.JMI.12.S2.S22016)
- [5] Hyun, S., **Lee, S.** Choi, I., Shin, C. & Cho, S. (2025). Asymmetric scatter kernel estimation neural network for digital breast tomosynthesis. *Journal of Medical Imaging*, 12(S2), S22008-S22008. doi: [10.1117/1.JMI.12.S2.S22008](https://doi.org/10.1117/1.JMI.12.S2.S22008)
- 2024** [6] Yun, S., **Lee, S.** Choi, D., Lee, T., & Cho, S. (2024). TMAA-net: Tensor-domain Multi-planal Anti-Aliasing network for Sparse-view CT Image Reconstruction. *Physics in Medicine & Biology*, 69(22), 225012. doi: [10.1088/1361-6560/ad8da2](https://doi.org/10.1088/1361-6560/ad8da2)
- [7] Kim, H., Kim, H., Eom, H., Choi, W., Chae, E., Shin, H., ... **Lee, S.** & Cho, S. (2024). Optimizing angular range in digital breast tomosynthesis: A phantom study investigating lesion detection across varied breast density and thickness. *Physica Medica*, 124, 103419. doi: [10.1016/j.ejmp.2024.103419](https://doi.org/10.1016/j.ejmp.2024.103419)
- 2023** [8] Kim, H.*, **Lee, S.***, ... & Cho, S. (2023). Homogenization of multi-institutional chest x-ray images in various data transformation schemes. *Journal of Medical Imaging*, 10(6), 061103. doi: [10.1117/1.JMI.10.6.061103](https://doi.org/10.1117/1.JMI.10.6.061103) (*: equal contributions)
- [9] Kim, H.*, Lee, H.*, **Lee, S.**, Choi, Y., ... & Cho, S. (2023). A feasibility study on deep-neural-network-based dose-neutral dual-energy digital breast tomosynthesis. *Medical Physics*, 50(2), 791-807. doi: [10.1002/mp.16071](https://doi.org/10.1002/mp.16071) (*: equal contributions)
- 2022** [10] **Lee, S.**, Kim, H., Lee, H., & Cho, S. (2022). Deep-learning-based projection-domain breast thickness estimation for shape-prior iterative image reconstruction in digital breast tomosynthesis. *Medical Physics*, 49(6), 3670-3682. doi: [10.1002/mp.15612](https://doi.org/10.1002/mp.15612)
- 2021** [11] Yun, S., Kim, Y., Kim, H., **Lee, S.**, Jeong, U., Lee, H., ... & Cho, S. (2021). Three-compartment-breast (3CB) prior-guided diffuse optical tomography based on dual-energy digital breast tomosynthesis (DBT). *Biomedical Optics Express*, 12(8), 4837-4851. doi: [10.1364/BOE.431244](https://doi.org/10.1364/BOE.431244)
- [12] Cho, S., **Lee, S.**, Lee, J., Lee, D., Kim, H., Ryu, J. H., ... & Cho, S. (2021). A novel low-dose dual-energy imaging method for a fast-rotating gantry-type CT scanner. *IEEE Transactions on Medical Imaging*, 40(3), 1007-1020. doi: [10.1109/TMI.2020.3044357](https://doi.org/10.1109/TMI.2020.3044357)

Conference Proceedings

- 2026** [1] **Lee, S.**, Davis, A., Kharat, S., Gonsalves, W., ... Baffour, F. & Gong, H. (2026, June) Physics-Informed Scalable CycleGAN for CT Reconstruction Kernel Conversion and Harmonization. In *CT Meeting 2026* [Poster]
- 2025** [2] **Lee, S.**, Hyun, S., & Cho, S. (2025, February). Evaluation of deep learning-based scatter correction in X-ray breast imaging: Across image domains and downsampling ratios. In *SPIE 2025 Medical Imaging*. SPIE, the international society for optics and photonics. [Poster]. doi: [10.1117/12.3048727](https://doi.org/10.1117/12.3048727)
- [3] Hyun, S., Choi, D., Yun, S., **Lee, S.**, & Cho, S. (2025, February). A diffusion-model-based dual-domain approach for CT metal artifact reduction. In *SPIE 2025 Medical Imaging*. SPIE, the International society for optics and photonics. [Poster]. doi: [10.1117/12.3046879](https://doi.org/10.1117/12.3046879)
- 2024** [4] **Lee, S.**, Hyun, S., Kim, D., & Cho, S. (2024, May). A YOLO-based learning lesion classifier of pre-exposure scan in digital breast tomosynthesis. In *17th International Workshop on Breast Imaging (IWBI2024)*. SPIE, the International Society for Optics and Photonics. [Oral]. doi: [10.1117/12.3025819](https://doi.org/10.1117/12.3025819)

- 2024** [5] Hyun, S., **Lee, S.**, Jeong, U., & Cho, S. (2024, May). Asymmetric scatter kernel superposition-inspired deep learning approach to estimate scatter in breast tomosynthesis. In *17th International Workshop on Breast Imaging (IWBI2024)*. SPIE, the International Society for Optics and Photonics. [Oral]. doi: [10.1117/12.3024774](https://doi.org/10.1117/12.3024774)
- [6] **Lee, S.**, Kim, H., Kim, H., & Cho, S. (2024, February). Predict Risk of Stereotactic Body Radiotherapy Induced Vertebral Compression Fracture Using Multi-Modal Deep Learning Network. In *SPIE 2024 Medical Imaging*. SPIE, the International society for optics and photonics. [Poster]. doi: [10.1117/12.3006647](https://doi.org/10.1117/12.3006647)
- 2022** [7] Kim, H.*, Lee, H.*, **Lee, S.**, Choi, Y., ... & Cho, S. (2022). A feasibility study of dual-energy digital breast tomosynthesis for three-compartment-breast imaging. In *SPIE 2022 Medical Imaging*. SPIE, the International society for optics and photonics. [Oral]. doi: [10.1117/12.2611606](https://doi.org/10.1117/12.2611606)
- 2020** [8] **Lee, S.**, Kim, H., Lee, H., Jeong, U., & Cho, S. (2020, May). Convolutional-neural-network based breast thickness correction in digital breast tomosynthesis. In *15th International Workshop on Breast Imaging (IWBI2020)* (Vol. 11513, p. 115131E). SPIE, the International Society for Optics and Photonics. [Oral]. doi: [10.1117/12.2560909](https://doi.org/10.1117/12.2560909)

Conference Presentations - International

- 2026** [1] **Lee, S.**, Davis, A., Kharat, S., Gonsalves, W., ... Baffour, F. & Gong, H. (2026, November) Scalable physics-informed CT image conversion: From intra-system training to cross-generation and cross-vendor generalization. In *2026 RSNA*. Radiological Society of North America. [Oral]
- [2] **Lee, S.**, Ferrero, A., Rajendran, K., ... Baffour, F. & Gong, H. (2026, November) Deep-learning-based trabecular morphometry for in situ vertebral microarchitecture assessment using clinical PCD-CT. In *2026 RSNA*. Radiological Society of North America. [Oral]
- [3] Hyun, S., **Lee, S.**, Kim, D., Lim, D., & Cho, S. (2026, November). Preliminary Study on Thick-Slab Image Generation of Digital Breast Tomosynthesis. In *2026 IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detector*. IEEE. [Poster]
- 2023** [4] **Lee, S.**, Hyun, S., & Cho, S. (2023, July). Enhancing of Scout Scan Using Diffusion Denoising in Digital Breast Tomosynthesis. In *AAPM 2023 65th Annual Meeting & Exhibition*. American Association of Physicists in Medicine. [Poster]
- [5] Choi, Y., **Lee, S.**, & Cho, S. (2023, July). Automatic Instant Teeth Segmentation of Panoramic Radiographs Using Multi-Frequency Processing. In *AAPM 2023 65th Annual Meeting & Exhibition*. American Association of Physicists in Medicine. [Oral, Presenter]
- [6] **Lee, S.**, Hyun, S., Kim, D., & Cho, S. (2023, January). Model observer analysis for suggestion of angular range in digital breast tomosynthesis based on patient breast type. In *International Forum on Medical Imaging in Asia (IFMIA) 2023*. IFMIA. [Oral]
- 2022** [7] Hyun, S., **Lee, S.**, Kim, H., & Cho, S. (2022, July). Deep-Unfolding-Network-Based Non-Blind Deblurring for Fast-Rotating Wide-Angle Digital Breast Tomosynthesis. In *AAPM 2022 64th Annual Meeting & Exhibition*. American Association of Physicists in Medicine. [Oral]
- 2021** [8] **Lee, S.**, Kim, H., Kim, H., & Cho, S. (2021, November). Multi-modal Deep-learning Based Prediction of Stereotactic Body Radiotherapy Induced Vertebral Compression Fracture. In *2021 RSNA*. Radiological Society of North America. [Oral]
- [9] **Lee, S.**, Kim, H. Sim, W., Cho, D., & Cho, S. (2021, September). Analysis of Deep Neural Network Performance Enhancement via Radiomic and Deep Feature Extraction. In *The 9th Korea-Japan Joint Meeting on Medical Physics*. KJMP. [Oral]
- [10] Kim, H., **Lee, S.**, Kim, H., & Cho, S. (2021, September). Prediction of Vertebral Compression Fracture after Stereotactic Body Radiotherapy using Multi-modal Network. In *The 9th Korea-Japan Joint Meeting on Medical Physics*. KJMP. [Oral]
- [11] **Lee, S.**, Kim, H., Lee, H., & Cho, S. (2021, July). Breast thickness map estimation and its associated correction in DBT imaging. In *AAPM 2021 63rd Annual Meeting & Exhibition*. American Association of Physicists in Medicine. [Poster]
- [12] Kim, H., **Lee, S.**, Sim, W., Cho, D., & Cho, S. (2021, July). Multi-frequency-based CXR data normalization for deep-neural-network classifier. In *AAPM 2021 63rd Annual Meeting & Exhibition*. American Association of Physicists in Medicine. [Poster]
- 2019** [13] Cho, S., **Lee, S.**, Lee, J., & Cho, S. (2019, October). Beam-filter-based dual-energy CT imaging by use of sinogram streaking. In *IEEE Nuclear Science Symposium and Medical Imaging Conference*. IEEE NSS/MIC. [Poster]

Conference Presentations - Domestic (Korea)

- 2024** [1] **Lee, S.**, & Cho, S. (2024) Comparative Analysis of Deep Learning-Based Scatter Correction Across Various Domains and Downsampling Ratios, In *2024 Korean Society of Radiation Industry*. Korean Society of Radiation Industry. [Poster]
- 2023** [2] **Lee, S.**, Hyun, S., Kim, D., & Cho, S. (2023) You Only Look Pre-shot: A YOLO-based lesion detection of pre-shot exposure data in tomosynthesis, In *2023 Korean Society of Radiation Industry*. Korean Society of Radiation Industry. [Poster]
- [3] **Lee, S.**, Hyun, S., & Cho, S. (2023) Denoising of pre-shot images using the diffusion model for digital breast tomosynthesis. In *65th Korean Society of Medical Physics*. Korean Society of Medical Physics. [Poster]
- 2022** [4] **Lee, S.**, Hyun, S., & Cho, S. (2022) Statistical evaluation of the computational reader in multi-angle digital breast tomosynthesis system for parameter suggestion. In *64th Korean Society of Medical Physics*. Korean Society of Medical Physics. [Oral]
- 2020** [5] **Lee, S.**, Kim, H., & Cho, S. (2020) Breast shape reconstruction during digital breast tomosynthesis based on discrete algebraic reconstruction technique. In *60th Korean Society of Medical Physics*. Korean Society of Medical Physics. [Oral]

Patents

- [1] **Lee, S.**, Baffour, F., Gong, H., & McCollough, C. H. *Systems and Method for Physics-Informed CT Image Kernel Conversion Using Deep Learning*. U.S. Patent Application No. **64/078,116**, filed May 29, 2026.
- [2] Cho, S., **Lee, S.**, Hyun, S., & Kim, D. *System of Determining Patient-Specific Angular Range for Digital Breast Tomosynthesis and Method Thereof*. Korean Patent Application No. **10-2024-0168384**, filed Nov. 22, 2024; published May 29, 2026.

Professional Services

Ad-hoc Associate Editor , <i>Medical Physics</i>	2026 – Present
Reviewer , <i>IEEE Transactions on Radiation and Plasma Medical Sciences</i> (2 reviews)	2026 – Present
Reviewer , <i>Medical Physics</i> (13 reviews)	2023 – Present
Reviewer , <i>Physics in Medicine & Biology</i> (11 reviews)	2023 – Present
Reviewer , <i>Biomedical Physics & Engineering Express</i> (4 reviews)	2022 – Present
Reviewer , MICCAI 2026	2026
Reviewer , MICCAI 2025	2025

Honors and Awards

Trainee Research Prize 2026 Radiological Society of North America (RSNA) rsna.org/research/research-awards/trainee-research-prize	Nov. 2026
Nuclear Global Postdoctoral Fellowship 2025 Nuclear Global Postdoctoral Fellowship, Korea Nuclear International Cooperation Foundation (KONICOF). <i>Selected, declined due to institutional policy</i>	Aug. 2025
Best Presentation Award The 2024 Meeting for Emerging Researchers of the Korean Society of Radiation Industry	Jul. 2024
Curie Student Award Korea Nuclear Society	Oct. 2023
Best Presentation Award International Forum on Medical Imaging in Asia (IFMIA)	Jan. 2023
Young Investigator Award The 9th Korea-Japan Joint Meeting on Medical Physics	Sep. 2021
Best Student Research Award Korean Society of Medical Physics	Sep. 2020
Lee Byong Whi Scholarship Top Incoming Student Scholarship, Dept. of Nuclear and Quantum Eng., KAIST	Mar. 2019
National Excellence Scholarship (Natural Sciences and Engineering) Ministry of Science and ICT, Republic of Korea	2017 - 2018

Teaching Experience

Special Lecturer - NQe599 Tomographic Image Reconstruction, KAIST	Spring 2025
Teaching Assistant - NQe471 Experiments in Quantum Engineering, KAIST	Spring 2023
Undergraduate Research Mentoring - Mentoring Undergraduate Individual Research, KAIST	Summer 2022
Counselling Assistant - Nuclear and Quantum Engineering Department, KAIST	2022
Teaching Assistant - NQe471 Experiments in Quantum Engineering, KAIST	Spring 2021
Special Lecturer (English) - NQe503 Radiation Science, Technology, and Applications, KAIST	Spring 2021
Teaching Assistant (English) - NQe201 Fundamentals of Nuclear and Quantum Science, KAIST	Spring 2019

Invited Talks

IEEE NPSS Seoul Chapter "Advancing digital breast tomosynthesis imaging techniques with machine learning"	Feb. 2025
---	-----------

Languages

Korean	Native
English	Fluent (TOEFL iBT: 101/120)
Japanese	Intermediate Listener and Speaker, Novice Reading and Writing

Computer Skills

Programming	Python, C++, CUDA, JAVA
Applications	MATLAB, IDL/ENVI, MySQL
Platforms	MS Windows, Linux (Ubuntu, OpenSUSE, CentOS)

Miscellaneous

Award-winning entry Nubzuki's Shinsegae Adventure Art Exhibition, KAIST Art Museum. art.kaist.ac.kr/doc/ko/pastExhibit.do?docSeq=2691	2023
Grand prize 'The KAIST I Loved, the KAIST That Loved Me' Writing Competition, KAIST School of Digital Humanities and Computational Social Sciences. ISBN 978-89-522-4000-2-43410	2018

References

Dr. Seungryong Cho	Professor, Dept. of Nuclear and Quantum Engineering, KAIST, Daejeon, Korea scho@kaist.ac.kr
Dr. Hyeon Lee	Assistant Professor, Dept. of Diagnostic Radiology and Centre of Cancer Medicine, University of Hong Kong, Hong Kong hlee01@hku.hk
Dr. Hao Gong	Assistant Professor, Radiology; Senior Associate Consultant I, Dept. of Artificial Intelligence and Informatics, Mayo Clinic, Rochester, MN, USA gong.hao@mayo.edu